Supplemental Material for "Local-resolvent formulation and assessment of linear-scaling Kohn–Sham calculations"

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This Supplemental Material gives the data behind statements in the main text that are not shown there in full. All numbers are generated by the scripts listed in Section S9 from the files in `results/`. Lemma, Proposition, Theorem and Remark numbers refer to the main text.

S1. KOHN–SHAM REFERENCE SOLVERS

Dense solvers. We compared three LAPACK paths for the dense generalized eigenproblem, and two ways of building the density from the eigenvectors (Table 1). The paths were `dsygvx` (bisection and inverse iteration for the lowest $N/2+1$ pairs), `dsygvd` (divide and conquer, all pairs), and Cholesky reduction with `dsyevr` (MRRR, the lowest $N/2+1$ pairs). The density was built either as the orbital sum $2\sum_i |\sum_\mu X_{\boldsymbol{r}\mu} C_{\mu i}|^2$ or by projecting $P = 2CC^{\mathsf{T}}$ (formed with `dsyrk`) through the sparse basis. The six variants give the same density. The automated test asserts agreement to 10^{-10}, and the observed differences are of order 10^{-15}.

For 432 electrons and more the eigensolver takes more than 84% of the time. For 128 electrons it takes about 40%, and building the density takes 8% (orbital sum) to 19% (density matrix). `dsygvx` and `dsyevr` are equally fast and `dsygvd` is slower; we use `dsygvx` with the orbital sum. In 1D, bisection (`stebz`) with inverse iteration was faster than MRRR (`stemr`) for the occupied states of dimerized chains (Table 2).

Warm-started LOBPCG. As an iterative reference we used LOBPCG on the pencil (H, S), with $N/2+24$ vectors, a block-Jacobi preconditioner $(H-\sigma S)^{-1}$ with one block per molecule and σ 0.5 Ha below the lowest eigenvalue, and starting vectors from the previous SCF step. The first step is solved densely. Table 3 shows that it is slower than dense diagonalization at all sizes tested, with up to 63 LOBPCG iterations per SCF step. At 1024 electrons a warm LOBPCG step took about 46 s, against 38 s for a dense step.

Table 1. 3D dense KS density build (s, single thread, best of 2–3): eigensolver time, density time and total.

N	N_b	driver	density	eigensolver	density	total
128	768	`gvx`	orbitals	0.06	0.01	0.15
128	768	`gvx`	DM	0.06	0.03	0.17
128	768	`gvd`	orbitals	0.11	0.01	0.20
128	768	`gvd`	DM	0.11	0.03	0.21
128	768	`evr`	orbitals	0.06	0.01	0.15
128	768	`evr`	DM	0.06	0.03	0.17
432	2592	`gvx`	orbitals	2.44	0.15	2.89
432	2592	`gvx`	DM	2.45	0.16	2.91
432	2592	`gvd`	orbitals	4.21	0.15	4.67
432	2592	`gvd`	DM	4.21	0.16	4.67
432	2592	`evr`	orbitals	2.45	0.15	2.91
432	2592	`evr`	DM	2.45	0.16	2.91
1024	6144	`gvx`	orbitals	36.19	1.25	38.33
1024	6144	`gvx`	DM	36.11	0.79	37.98
1024	6144	`gvd`	orbitals	60.92	1.18	63.44
1024	6144	`gvd`	DM	60.57	0.79	62.56
1024	6144	`evr`	orbitals	36.40	1.18	38.86
1024	6144	`evr`	DM	36.47	0.79	38.45

Table 2. 1D tridiagonal KS solvers for the occupied states (s, best of 2).

N	N_g	`stebz`	`stemr`
256	5763	0.233	0.878
512	11395	1.255	3.117
1024	22659	8.188	10.897

Table 3. Complete SCF calculations with warm-started LOBPCG and with dense KS: tolerance of LOBPCG, SCF iterations, energy difference to dense KS per electron, wall times (s), and the largest number of LOBPCG iterations in one SCF step.

N	tol	SCF it.	$\lvert\Delta E\rvert/N$	LOBPCG wall	dense wall	max. LOBPCG it.
128	1e-05	11	5.5×10^{-11}	3.2	2.0	63
128	1e-07	11	8.4×10^{-14}	4.3	2.0	63
432	1e-05	11	9.0×10^{-11}	43.8	34.3	63
432	1e-07	11	4.6×10^{-13}	66.9	34.3	63

S2. TIMING METHODOLOGY, REPEATABILITY AND THREADS

All timings are wall-clock times measured with `time.perf_counter` on an 8-core workstation (Linux, OpenBLAS 0.3.26).

Complete SCF calculations. Table 4 lists every run behind the SCF wall times of the main text (protocol: main text, Computational details, Timing). Both methods needed the same number of SCF iterations at each size. The 2000-electron LRes-DFT value is a single run.

Table 4. Wall time (s) of every complete SCF calculation, in run order (single thread, idle machine).

N	KS	LRes-DFT	SCF it.
54	0.5, 0.6, 0.5	3.2, 3.3, 3.2	11
128	2.0, 1.9, 1.9	8.6, 8.9, 8.4	11
250	7.4, 7.2, 7.1	25.2, 24.7, 25.0	11
432	35.2, 34.6, 33.9	46.5, 46.0, 44.6	11
686	136.5, 132.0, 131.4	65.4, 66.4, 65.8	11
1024	445.1, 441.2, 442.9	126.0, 125.0, 125.4	11
2000	–	272.1	11

Density builds. For LRes-DFT, the warm step reuses μ and E_{lo} from a previous call and computes only window eigenpairs below $c + 1.5R$. The cold step computes all window eigenvalues, determines μ, and then builds the density. The window vectors on the owned points are stored, so re-solving μ in a warm step needs no second eigensolver pass. The density-build series of the main text was measured on a shared machine at a load average near 1 (best of 2–3). Table 5 shows a separate series of five repetitions taken earlier under heavier load. There the standard deviations reached 12% (432 electrons) and 20% (686 electrons) for KS and up to 10% for LRes-DFT. This is why the main text uses complete SCF calculations on an idle machine for the headline comparison. Table 6 gives shared-memory timings at 432 electrons (load average 1–3): OpenMP over windows for LRes-DFT, and threaded OpenBLAS for KS.

Table 5. Repeatability under load: mean and standard deviation (s) of five density builds.

N	KS mean	KS s.d.	LRes-DFT mean	LRes-DFT s.d.
250	0.620	0.004	1.647	0.010
432	3.455	0.415	2.834	0.122
686	13.510	2.679	4.628	0.447

Table 6. Shared-memory threads, 432 electrons (s, best of 3).

threads	LRes-DFT warm step	KS	load average
1	2.66	2.90	0.8
2	1.54	2.26	1.0
4	1.16	2.59	1.5
8	0.97	2.72	2.6

S3. MEMORY

The peak resident memory of the solver step was measured by resetting the peak counter of the process after the basis, grid, Hamiltonian and window index data had been built (`/proc/self/clear_refs`). The high-water mark (`VmHWM`) was then read after the step (Table 8). The LRes-DFT process holds an additional 86 MB of window index data at 1024 electrons, which is not included in the step memory. The LRes-DFT step memory grows faster than linearly in this range because it includes the stored window vectors and per-window work arrays.

S4. BASIS SET AND WINDOW ERROR

Table 9 compares three bases:

- compact: five functions per atom, s exponents 1.0 and 0.25, p exponent 0.6 bohr^{-2}, cutoff 4.5 bohr;
- the basis used in the main text: s exponents 1.2, 0.4 and 0.15, p exponent 0.5, cutoff 4.5 bohr;
- diffuse: s exponents 1.0, 0.3 and 0.12, p exponent 0.5, cutoff 5.5 bohr.

The basis error is the difference between the LCAO-KS and real-space-grid KS energies for a 16-electron crystal. The window error is the LRes-DFT–KS difference at the self-consistent potential of a 250-electron crystal. A more complete, more diffuse basis lowers the basis error but raises the window error.

S5. POLE RULES: MARGIN, ERROR BOUND, CLOSING GAP AND OCCUPATION RANGE (1D)

Table 10 and Figure S1(b) show the density error of the $(H_2)_{16}$ chain as a function of the margin δ, with the rate ρ of Theorem 1. The per-state bound ρ^M in that table is multiplied by an estimate $2(N_{\mathrm{occ}}+5)/N$, so it is only indicative. The full bound of Theorem 1, evaluated exactly as stated, is given in Table 11 and plotted in Figure 3(a) of the main text. Table 12 gives the number of poles needed for uniform H chains whose gap closes with length. Table 13 gives the range of the occupations of the Gauss–Legendre and Zolotarev rules at the order needed for 10^{-6} accuracy. The half-step rule stays within $[0, 1]$ in all cases.

S6. GRID REFINEMENT AGAINST A CHEBYSHEV EXPANSION

For an $(H_2)_4$ chain, Table 14 compares the number of poles M of the half-step rule with the degree of a Chebyshev interpolant of a Fermi function needed for a density error of 10^{-6} per electron. The smearing of the Fermi function

Table 7. 1D cost data (s): LRes-DFT with the exact recursion including the search for μ from scratch, one LRes-DFT evaluation at fixed μ, the window variant, the tridiagonal KS solver and dense KS, and the window error per electron.

N	N_g	LRes-DFT	LRes-DFT, fixed μ	windows	KS (stebz)	KS (dense)	window error
16	483	0.014	0.0004	0.035	0.001	0.02	0
32	835	0.020	0.0008	0.064	0.004	0.07	4.1×10^{-8}
64	1539	0.058	0.0013	0.265	0.021	1.83	1.3×10^{-7}
128	2947	0.127	0.0027	0.328	0.056	10.58	1.5×10^{-7}
256	5763	0.361	0.0086	1.211	0.621	80.45	1.5×10^{-7}
512	11395	0.729	0.0167	2.525	3.190	–	1.6×10^{-7}
1024	22659	0.858	0.0217	3.437	22.733	–	1.6×10^{-7}
2048	45187	1.749	0.0449	5.661	216.198	–	1.6×10^{-7}
4096	90243	4.281	0.1164	18.224	–	–	1.6×10^{-7}
8192	180355	9.762	0.2604	nan	–	–	–

Table 8. Memory of one density build (MB): resident memory before the step and additional peak memory of the step.

N	N_b	method	before	additional peak
128	768	LR-DFT	168	21
128	768	KS	165	21
250	1500	LR-DFT	251	39
250	1500	KS	248	28
432	2592	LR-DFT	341	83
432	2592	KS	338	112
686	4116	LR-DFT	494	130
686	4116	KS	491	271
1024	6144	LR-DFT	666	251
1024	6144	KS	580	676

was chosen so that its own error is below the target, which requires a small temperature and a high degree. A damped polynomial approximant of the gapped step would need a lower degree. A polynomial term also costs one matrix–vector product, while a pole costs one linear solve. The comparison therefore shows only that the pole count does not depend on the grid spacing while the polynomial degree does; it is not a cost comparison.

S7. POINTWISE PROBING ON A REAL-SPACE GRID

Table 15 documents the abandoned 3D route. For a two-molecule H_2 cell on a 0.6 bohr grid, we compare the probed diagonal of the exact density matrix, $\sum_{k \in c(\boldsymbol{r})} P_{\boldsymbol{r}k}$, with its true diagonal. Here $c(\boldsymbol{r})$ is the set of grid points with the same color as $\boldsymbol{r}$ in a cubic coloring of spacing d. The relative error is the summed absolute error divided by the summed diagonal.

S8. WINDOW DIAGNOSTICS AND SCF CONVERGENCE

Table 16 and Figure S1(c) summarize the eigenvalues of all windows. No window has a state inside the KS gap, and the lowest window eigenvalue is never below the lowest KS eigenvalue, as Lemma 4 requires.

Figure S1(a) shows SCF convergence histories. LRes-DFT converges at the same rate as KS until the change in density per electron reaches 10^{-9} (3D) to 10^{-10} (1D). Below this level the density changes slightly from step to step, because μ is determined only to the tolerance of its search, and different values of μ inside the gap give densities that differ at this level. All SCF results in the main text use convergence thresholds above this floor (10^{-6}–10^{-8} per

Table 9. Basis-set trade-off.

basis	basis error/N (Ha)	$\lambda_{\min}(S)$	R_w	n_W	$\|\Delta n\|_1/N$	$\|\Delta E_{\mathrm{band}}\|/N$
compact (5 per atom, $r_c = 4.5$ bohr)	0.0505	0.0252	8.5	190	1.2×10^{-5}	1.9×10^{-7}
compact (5 per atom, $r_c = 4.5$ bohr)	0.0505	0.0252	12.1	330	1.3×10^{-7}	8.5×10^{-9}
used (6 per atom, $r_c = 4.5$ bohr)	0.0188	0.0068	8.5	228	1.0×10^{-4}	2.0×10^{-6}
used (6 per atom, $r_c = 4.5$ bohr)	0.0188	0.0068	12.1	396	3.6×10^{-6}	1.5×10^{-7}
diffuse (6 per atom, $r_c = 5.5$ bohr)	0.0047	0.0040	8.5	228	5.9×10^{-4}	7.2×10^{-5}
diffuse (6 per atom, $r_c = 5.5$ bohr)	0.0047	0.0040	12.1	396	3.8×10^{-5}	4.8×10^{-6}

Table 10. Margin δ (Ha): rate ρ, measured density error per electron and indicative bound.

δ	M	ρ	error/N	bound/N
0.02	32	0.860	1.1×10^{-3}	1.1×10^{-2}
0.02	48	0.860	8.9×10^{-5}	9.5×10^{-4}
0.05	32	0.684	6.4×10^{-7}	6.8×10^{-6}
0.05	48	0.684	1.4×10^{-9}	1.5×10^{-8}
0.1	32	0.666	2.1×10^{-7}	3.0×10^{-6}
0.1	48	0.666	2.7×10^{-10}	4.5×10^{-9}
0.2	32	0.718	2.3×10^{-6}	3.2×10^{-5}
0.2	48	0.718	1.0×10^{-8}	1.6×10^{-7}
0.3	32	0.755	1.3×10^{-5}	1.7×10^{-4}
0.3	48	0.755	1.2×10^{-7}	1.9×10^{-6}
0.5	32	0.807	1.1×10^{-4}	1.4×10^{-3}
0.5	48	0.807	3.1×10^{-6}	4.4×10^{-5}
1.0	32	0.873	1.6×10^{-3}	1.7×10^{-2}
1.0	48	0.873	1.6×10^{-4}	2.0×10^{-3}
2.0	32	0.925	1.4×10^{-2}	1.0×10^{-1}
2.0	48	0.925	3.3×10^{-3}	3.1×10^{-2}

electron in 3D). The 3D histories shown were taken before the window vectors were stored; the convergence rate is not affected by that change.

S9. VERIFICATION AND REPRODUCTION

Computer algebra. `cas_verification.py` performs 30 checks with SymPy and 40-digit mpmath arithmetic:

- Lemma 1, for $M = 2, \dots, 14$ and as a matrix identity, including the non-orthogonal version;
- the weight sum $\sum_j \zeta_j = 0$;
- Lemma 2: the Fermi–Dirac form in $\ln|u|$ and both bounds;
- Proposition 1(b): the band energy from g;
- the identity of Remark 2, symbolically and numerically;
- Proposition 2: Jacobi's formula and the derivative of W_M;
- the stationarity of G_M at self-consistency, in a three-site model;
- Proposition 3: $\partial\Omega_M/\partial\mu = -N_M$ for a rigidly moving circle;

Table 11. Theorem 1 for the $(H_2)_{16}$ chain ($\delta = 0.3$ Ha): total L_1 density error, error per electron, and the bound $2(N_{\mathrm{occ}} + N_{\mathrm{near}})\rho^M + 2N_g 2^{-M}$.

M	$\|n_M - n\|_1$	per electron	bound
8	5.7×10^{-1}	1.8×10^{-2}	1.2×10^{1}
16	4.6×10^{-2}	1.4×10^{-3}	5.6×10^{-1}
24	4.2×10^{-3}	1.3×10^{-4}	5.7×10^{-2}
32	4.0×10^{-4}	1.3×10^{-5}	6.0×10^{-3}
48	3.9×10^{-6}	1.2×10^{-7}	6.8×10^{-5}
64	4.0×10^{-8}	1.3×10^{-9}	7.6×10^{-7}
96	4.5×10^{-12}	1.4×10^{-13}	9.6×10^{-11}
128	3.0×10^{-15}	9.3×10^{-17}	1.2×10^{-14}

Table 12. Uniform H_n chains (spacing 2 bohr): gap and number of poles M needed for density errors of 10^{-4} and 10^{-6} per electron (half-step rule, $\delta = 0.3$ Ha).

n	gap (Ha)	M (10^{-4})	M (10^{-6})
10	0.095	50	82
20	0.052	80	136
40	0.027	134	238
80	0.014	228	426

- Lemma 3: the recursion, symbolically for a 5×5 matrix;
- Lemma 4: five random pencils;
- the decay roots of the uniform-potential resolvent;
- the LDA potentials as derivatives of the energy densities;
- the pseudopotential Fourier transform;
- the 1D exchange closed form against direct quadrature.

These checks verify the identities on explicit instances; the general proofs are in Appendix A of the main text.

Automated tests. The 25 tests in `tests/` check:

- the recursion against dense inversion;
- window convergence;
- the closed form and occupation bounds against eigendecomposition;
- the derivative identity by finite differences;
- recovery of the diagonal of h;
- the band energy against the eigenvalue sum;
- a harmonic oscillator;
- the exchange derivative;
- self-consistent LRes-DFT against KS for four 1D systems;
- 3D grid KS against dense diagonalization;
- LCAO density normalization;

Table 13. Occupation range of the Gauss–Legendre (GL) and Zolotarev rules at the order needed for 10^{-6}: minimum occupation and maximum minus one.

case	gap (Ha)	GL min	GL max−1	Zolotarev min	Zolotarev max−1
H10 chain	0.095	-2.1e-07	5.5e-12	-3.0e-07	2.3e-07
H20 chain	0.052	-1.2e-05	3.3e-06	-7.5e-07	7.2e-07
H40 chain	0.027	-2.4e-05	1.7e-05	-7.2e-07	7.2e-07
H80 chain	0.014	-8.0e-08	8.1e-08	-1.7e-06	1.7e-06
core Z=4	0.107	-5.7e-06	3.3e-07	-6.8e-07	6.0e-07
core Z=8	0.045	-8.0e-06	6.6e-06	-9.2e-07	8.4e-07

Table 14. Grid refinement at fixed gap.

Δx	N_g	W (Ha)	Δ (Ha)	M	Chebyshev degree
0.2	134	50	0.232	36	1792
0.1	267	200	0.233	36	3648
0.05	533	800	0.233	36	6144
0.025	1065	3200	0.233	36	12288

- exactness of LRes-DFT when the window covers the cell;
- 3D window convergence and self-consistency;
- agreement of the C++ and Python implementations;
- thread independence;
- agreement of the six KS solver variants;
- Lemma 4 on actual windows;
- agreement of the stored-vector re-solution of μ with a full second pass.

Scripts. The data were produced by the following scripts:

- `benchmarks/accuracy.py`: exact two-electron and 1D KS comparisons;
- `benchmarks/scaling.py`: 1D cost;
- `benchmarks/extra_checks.py`: grid refinement and grand-canonical forces;
- `benchmarks/three_d.py`: 3D accuracy, window and cost;
- `benchmarks/referee_checks.py`, parts R1–R18:
  - DC comparison, compression, amorphous system, basis, margin, SCF histories;
  - repeatability, memory, closing gap, pole rules, forces;
  - window diagnostics and the Theorem 1 bound;
  - occupation range, probing, LOBPCG, fixed-N forces and threads;
- `benchmarks/scf3d_large.py`: SCF up to 1024 electrons;
- `benchmarks/ks_drivers.py`: KS solver comparison.

Figures and tables are generated by `manuscript/make_figures.py` and `manuscript/make_tables.py`. The C++ kernels are built with `build_cpp.sh`.

Table 15. Relative error of the probed density-matrix diagonal.

probes	spacing (bohr)	relative error
8	1.2	70.97
125	3.0	3.66
1000	6.0	0.07

Table 16. Window spectra at the self-consistent KS potential (250 electrons).

system	R_w	HOMO	LUMO	window states in gap	lowest KS	lowest window
crystal	8.5	-0.166	0.122	0	-0.2515	-0.2448
crystal	12.1	-0.166	0.122	0	-0.2515	-0.2472
amorphous	8.5	-0.125	0.104	0	-0.2933	-0.2918
amorphous	12.1	-0.125	0.104	0	-0.2933	-0.2928

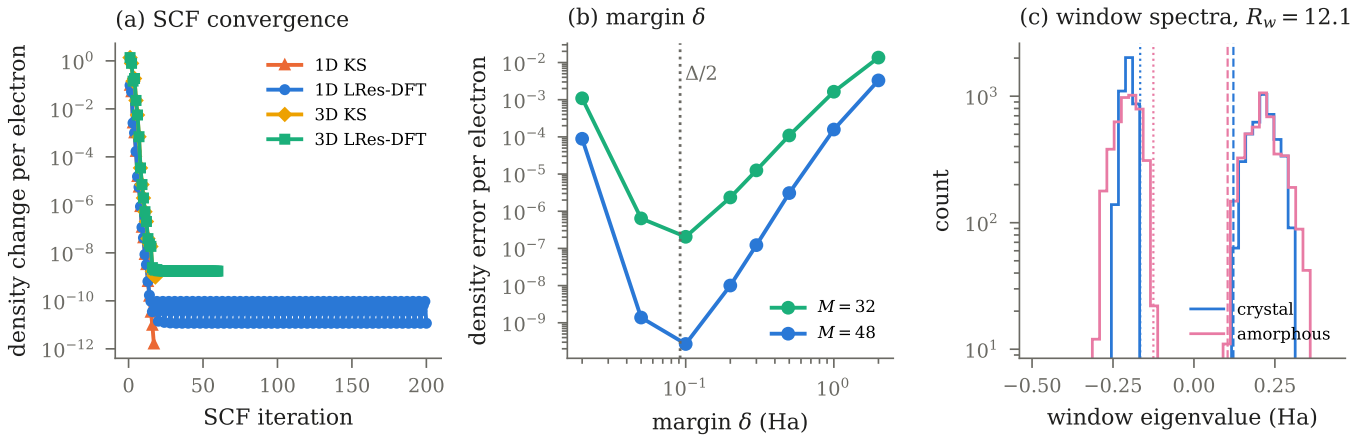


Figure S1. (a) SCF convergence of KS and LRes-DFT for a 1D $(H_2)_{16}$ chain and a 3D crystal with 128 electrons. (b) Density error per electron of the 1D chain versus the margin δ for $M = 32$ and 48; the dotted line marks half the gap. (c) Histogram of window eigenvalues at $R_w = 12.1$ bohr for the crystal and the amorphous arrangement; dotted and dashed lines mark the KS HOMO and LUMO.